

Sam Thompson

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EDUCATION

University of Cambridge (Trinity College)

Cambridge, UK

PhD in Engineering

Oct 2024 – Present

- Research on embodied conversational agents in social VR, focusing on presence, embodiment, and user behaviour
- Hands-on prototyping and development, experimental design and statistical analysis of user studies
- Funded by the Woolf Fisher Trust and Cambridge Trust
- Co-President, Cambridge University VR/AR Society
- Co-Founder, RealityX Hackathon

University of Auckland

Auckland, New Zealand

Bachelor of Advanced Science (Honours), Computer Science

Mar 2020 – Nov 2023

- Graduated with First Class Honours
- President, University of Auckland VR Club

EXPERIENCE

University of Auckland

Auckland, New Zealand

Research Assistant (HCI / XR & Software Engineering) and Graduate Teaching Assistant

Mar 2020 – Jul 2024

- Built various XR applications for experimental studies within the Computer Graphics group
- Contributed as a software engineer to a flood-resilient digital twin, focused on simulation and visualization. Worked with databases, geospatial data and defining an API
- Worked across research and engineering contexts to translate research goals into functional XR systems
- Taught parts of undergraduate and postgraduate courses in ML, theoretical CS and computing education
- Collaborated with interdisciplinary teams across the University of Auckland and the University of Canterbury

Intercollegiate XR (ICXR)

International

President

Feb 2025 – Feb 2026

- Led an international XR community and executive team, connecting students and early-career researchers

PROJECTS

Embodied Conversational Agents | *OpenAI Realtime API, C#, Unity*

October 2024 – Present

- Developed an embodied conversational agent (ECA), capable of interacting in real-time, and in standalone, mobile VR headsets

Redirected Walking | *Unity, C#, Blender, Android, Research*

2023 – 2024

- Developed a virtual environment prototype to enable natural walking in XR within a limited physical space

Stereoscopic Super-sampling for XR Devices | *Python, Unity, C#*

2023

- Supported development for a model to reconstruct XR views via neural upscaling
- Developed a custom dataset of colour, depth and motion vector information from stereoscopic views
- Compared results with other state-of-the-art models

TECHNICAL SKILLS

Languages: C#, Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS

Frameworks: Pytorch, p5.js, Processing, .NET, MonoBehaviour, OpenXR, BasisVR, Meta XR SDK

Developer Tools: Git, Unity, Blender, Substance Painter, Visual Studio, PyCharm, IntelliJ, Eclipse, Docker

Libraries: pandas, NumPy, Matplotlib